

Positioning Composite — Normalization Versions

Five interchangeable composite methodologies on one signal: the current triple-normalization baseline, a single-standardization Kaiser-PCA, the CFTC/Street best-practice, a single-pass EWMA/52w, and a two-dimension curve read. Built concurrently against one fixed data interface; the production default is unchanged.

1. Why and what

The audit showed the live composite is ~13 months slow because normalization is a triple pass (rolling-3y-z of the level → StandardScaler → PCA → rolling-3y-z of PC1). This note rebuilds the composite five ways — each a self-contained module in `positioning/versions/` coded against a fixed `base.py` interface (shared inputs: DV01-weighted per-category net positions, open interest, curve books, yields; shared normalizers), so they are directly comparable and *selectable* via `get_composite(version)`. `positioning/core` and the live `positioning_score.json` are untouched. Sign convention throughout: positive = crowded net long duration (curve: + = steeper).

2. The five versions

V0 — Production (baseline). Rolling-3y-z features → PCA (fit 2006–19, 2 fixed PCs) → rolling-3y-z of PC1. Latest $z = +2.49$.

V1 — Kaiser-PCA (single standardization, eigenvalue > 1). *The requested fix.* Raw per-category net-DV01 levels, standardized *once* (StandardScaler fit on the 2006–19 train slice, projected forward), PCA on the same slice, retain PCs with **eigenvalue > 1 (Kaiser)**; composite = PC1 read straight off the projection — *no* rolling re-z. Result: eigenvalues [2.05, 0.91, 0.04], so **one PC survives** (PC1 = 68.3% of variance; PC2 at 0.91 just misses Kaiser, so no separate curve factor here). Removing both rolling passes is exactly what kills the lag. Latest = +7.71 — legitimately large because PC1 is unit-variance on the *pre-2020* window and today's book is a multi-year extreme *outside* that window; V1 does not re-normalize that away (unlike V0). Read V1 as “distance from the pre-2020 normal”, not as a rolling z. (A documented `raw_cov` sub-variant — PCA on the unscaled covariance — is dominated by the largest-variance series, asset managers, confirming why standardizing first matters.)

V2 — CFTC / Street best-practice. The industry method: on the DV01/10y-equivalent buy-side net (the OFR convention), a bounded **COT-index** = mean of the 52w and 3y min-max stochastics (0–100, crowd > 80), with an unbounded `street_z` = $0.5 z_{52} + 0.3 z_{\%OI} + 0.2 z_{flow}$ in support (52w z; net-%-of-open-interest z; 13w flow z). Latest COT-index = 85.5 (crowded); sub-readings $cot_{52} = 81.6$, $cot_{156} = 89.4$, $z_{52} = 1.39$, $z_{\%OI} = 1.48$, $z_{flow} = -0.64$, `street_z` = 1.01.

V3 — EWMA / 52w single-pass. The simplest low-lag composite: $0.7 \times \text{EWMA-z (half-life 26w)} + 0.3 \times 13\text{w-flow z}$ of the buy-side level. One standardization, no PCA. Latest = +0.61 ($z_{52} = 1.39$, EWMA-z = 1.14) — already easing because flow turned negative.

V4 — Curve, two dimensions. Splits the steepener/flattener read into **book tilt** (raw front–long DV01, currently ≈ -2.6 \$mm/bp = flat) vs **momentum** (EWMA-z, +1.42 steepener-direction); plus a PCA slope factor (–1.40; its eigenvalue $0.22 < 1$, so it is a diagnostic, not Kaiser-retained) and a COT-index of the curve book (95.2, near its 52w top). Resolves the earlier “+1.46 crowded steepener” ambiguity: the *book* is flat; only the *momentum* is steepening.

3. Responsiveness — the whole point

Same underlying signal, when did each first flag “crowded” in the current build-up:

Version	Metric	Latest	First “crowded” flag	Lead vs production
V0 Production	3y double-z	+2.49	2026-01-20	—
V1 Kaiser-PCA	single-std PC1	+7.71	2024-06-04	+85 wks (~20m)
V3 EWMA/52w	EWMA-z + flow	+0.61	2024-08-13	+75 wks (~17m)
V2 CFTC/Street	COT-index (0–100)	85.5	2024-12-24	+56 wks (~13m)
V4 Curve	momentum (curve, not duration)	+1.42	2026-03-03	n/a (different signal)

Every duration-crowding alternative flagged the crowd **13–20 months** before the production composite. The tell: V1/V3 have already *peaked and are rolling off* while V0 is only now printing its high — the 3y double-z is a rear-view mirror. (V4 is the curve dimension; its “lead” vs a duration metric is not comparable.)

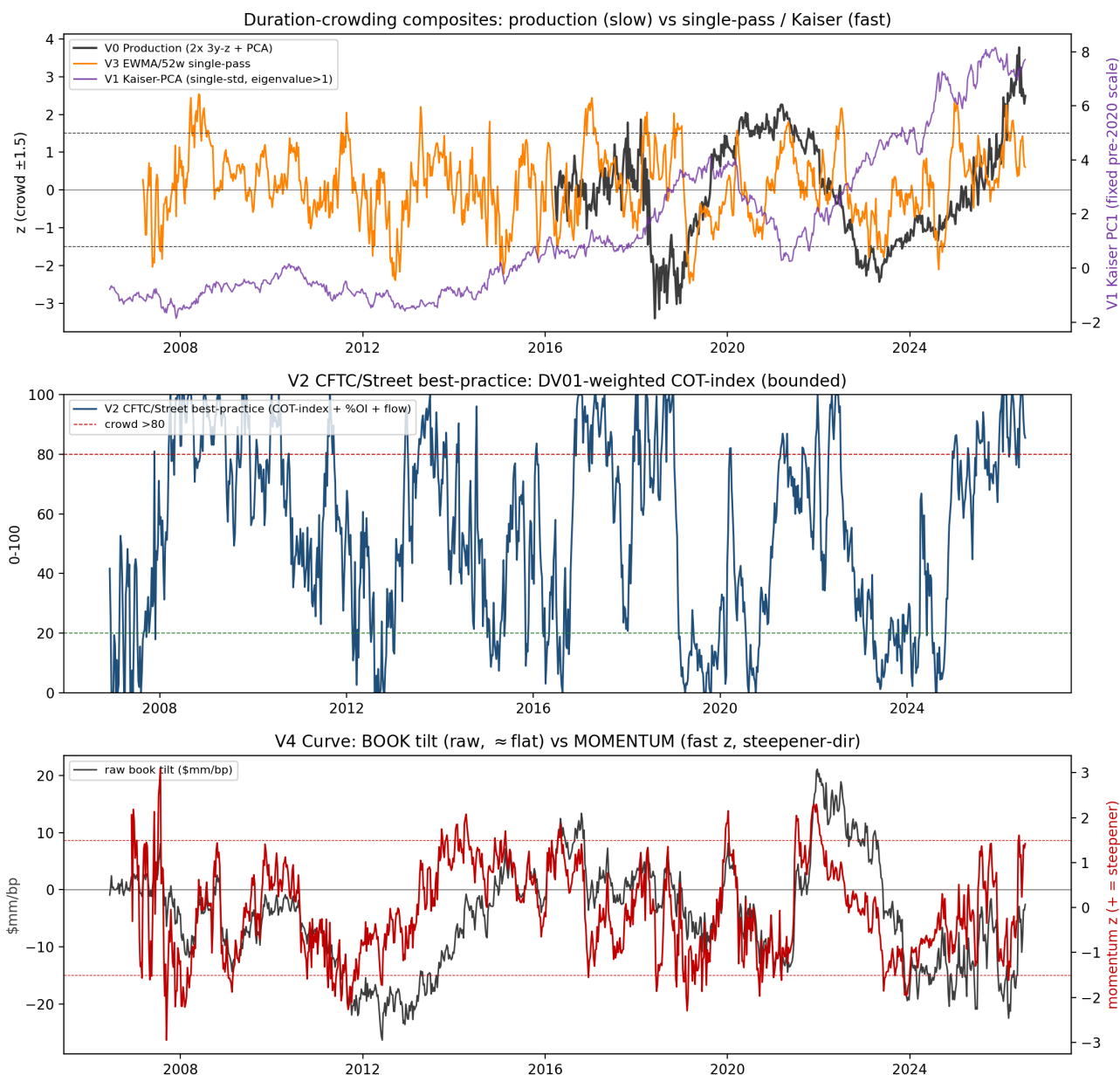


Figure 1: *

Fig 1. Top: duration-crowding composites — production (grey, slow) vs EWMA/52w (orange) with Kaiser-PC1 on the right axis (fixed pre-2020 scale, hence the upward drift). Middle: V2 bounded COT-index (saturates fast, crowd >80). Bottom: V4 curve — raw book tilt ($\approx$ flat now) vs momentum (steepener-direction).

4. Recommendation

Keep the tool's DV01 weighting and PCA (already OFR / Litterman-Scheinkman best practice); change only the normalization, and offer the versions as selectable:

- **Headline, magnitude-comparable, fast:** V3 (single-pass EWMA/52w) — same $\pm z$ scale as today, ~ 17 months faster, trivially simple.
- **Desk-friendly bounded gauge:** V2 COT-index (0–100, crowd >80) as a secondary read — matches how the Street quotes crowding; accept that it saturates.
- **Factor view:** V1 Kaiser-PCA for the “distance from pre-2020 normal” — most principled on the double- z fix, but present its magnitude with the fixed-scale caveat.
- **Structural context:** keep V0 3y as a slow band, not the headline.
- **Curve:** adopt V4's two-series read (book tilt + momentum) everywhere the single z misled.

Honesty note. These are *normalization* choices — they change *when* a given book is called crowded, not whether crowding predicts returns (which the earlier study showed is weak: CIs include 0.5). A faster flag is a better *descriptor*, not new alpha. Production default remains V0 until a switch is chosen; `get_composite("v1".."v4")` exposes each.

Reproduce: `python -m positioning.versions.compare (figures + versions_compare.json)`; each version self-runs via `python -m positioning.versions.vX`. Built concurrently by four agents against `positioning/versions/base.py`; `positioning/core` and `positioning_scores` unchanged. Methodology sources in the companion note `positioning_methodology_review.pdf`.